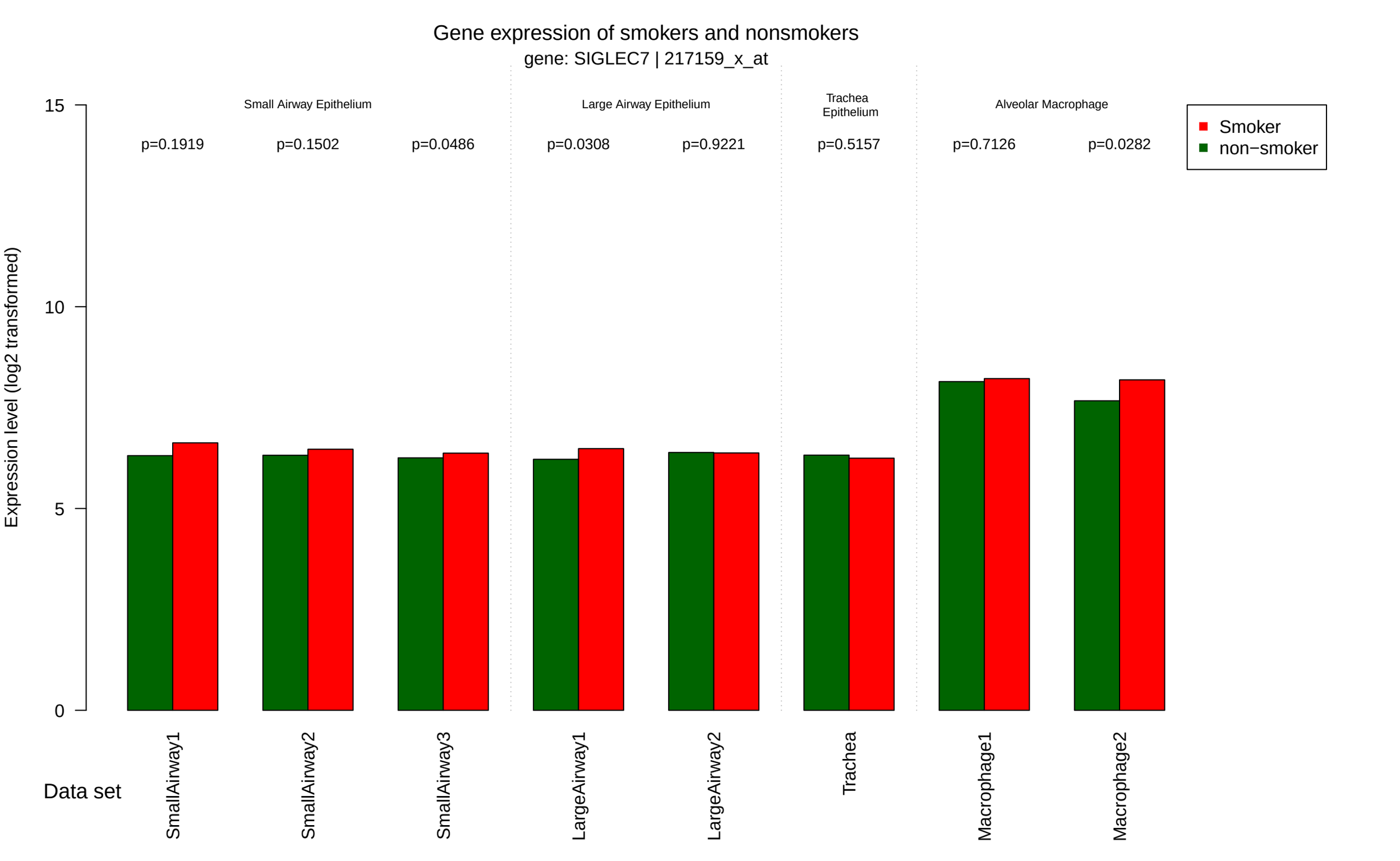
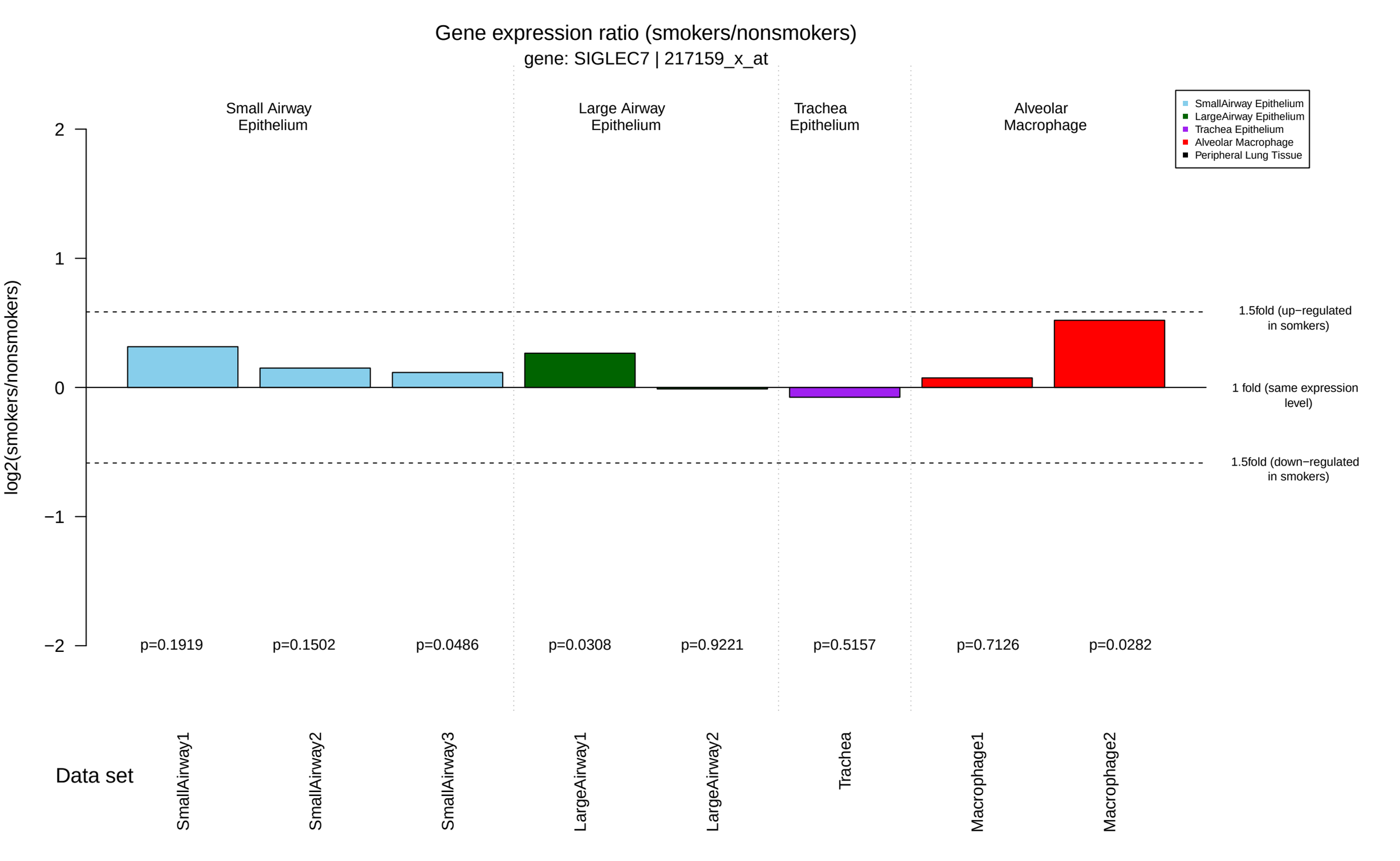
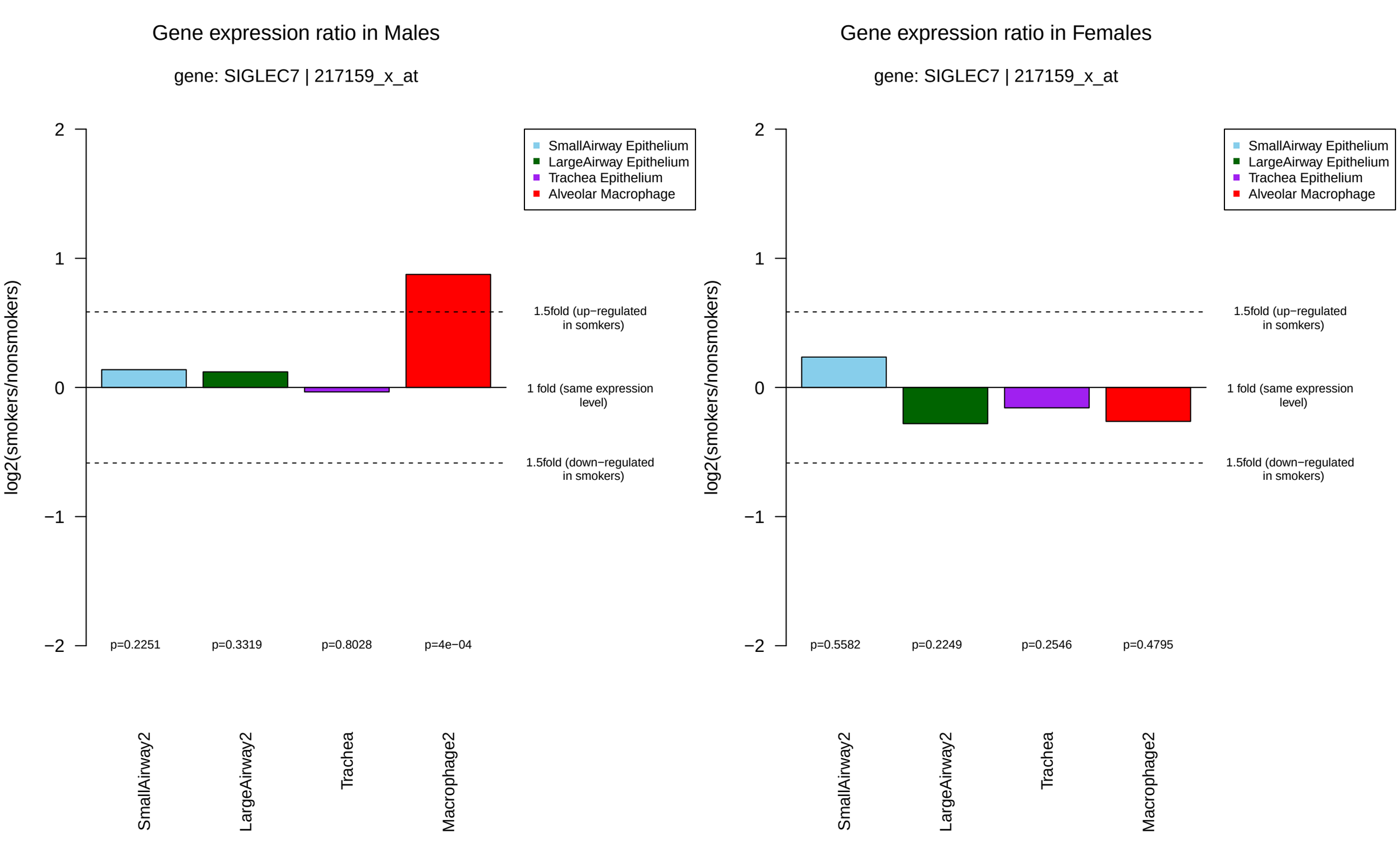
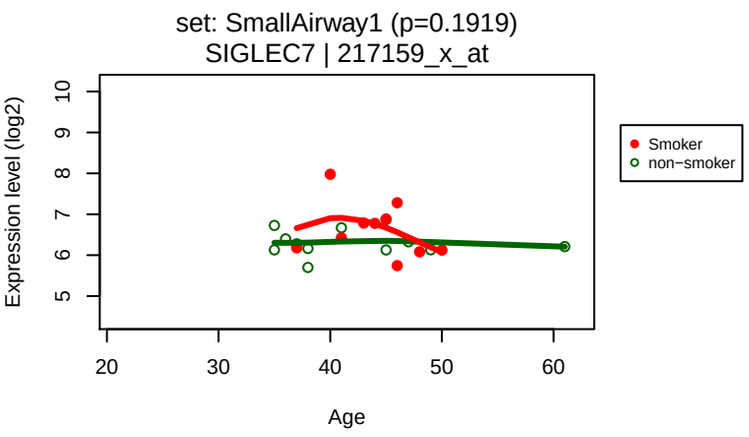
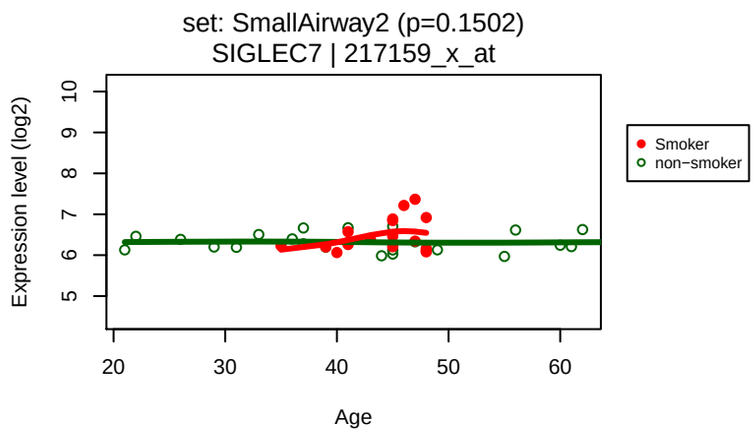
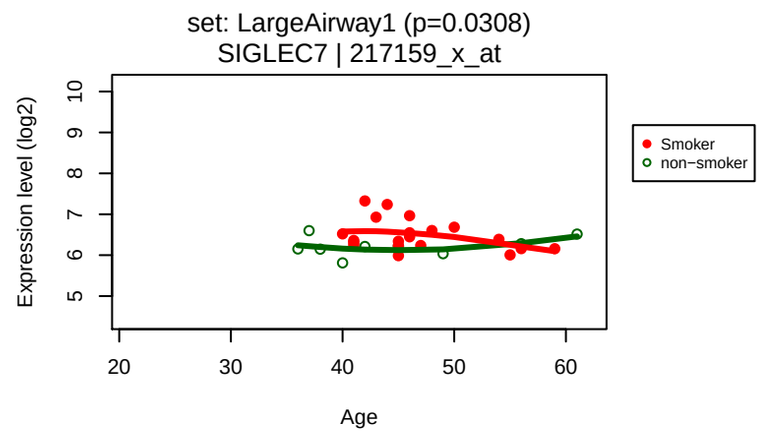

Gene expression of smokers and nonsmokers
gene: SIGLEC7 | 217159_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1919
p=0.1502
p=0.0486
p=0.0308
p=0.9221
p=0.5157
p=0.7126
p=0.0282
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

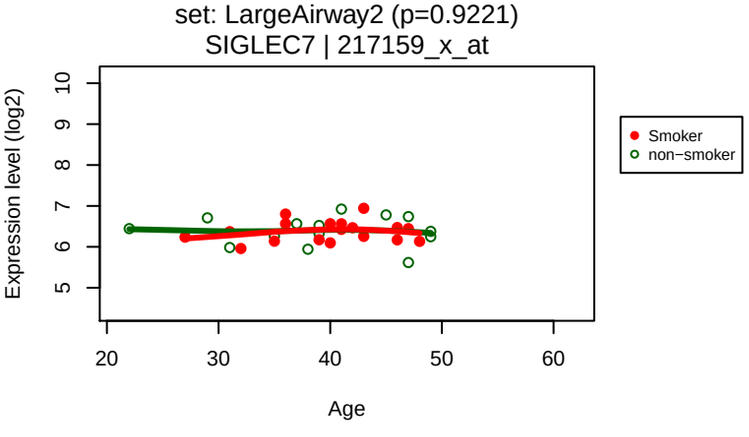

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC7 | 217159_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1919
p=0.1502
p=0.0486
p=0.0308
p=0.9221
p=0.5157
p=0.7126
p=0.0282
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

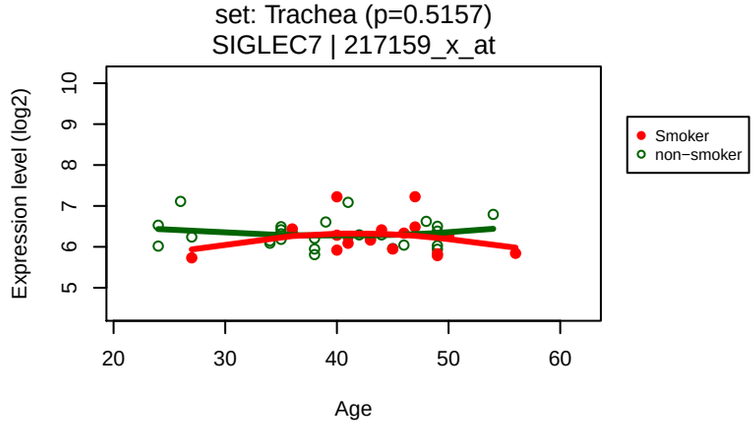

Gene expression ratio in Males
gene: SIGLEC7 | 217159_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2251
p=0.3319
p=0.8028
p=4e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC7 | 217159_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5582
p=0.2249
p=0.2546
p=0.4795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

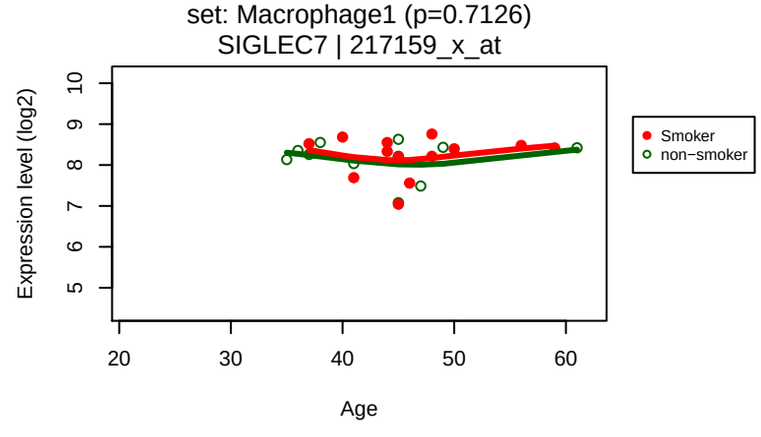

set: SmallAirway1 (p=0.1919)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker

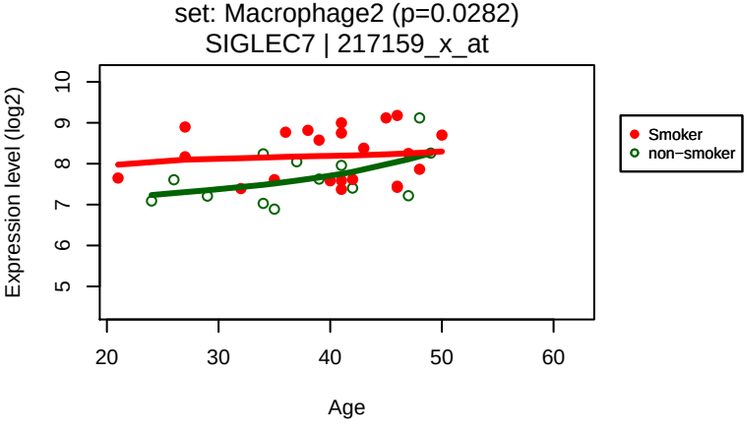

set: SmallAirway2 (p=0.1502)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker

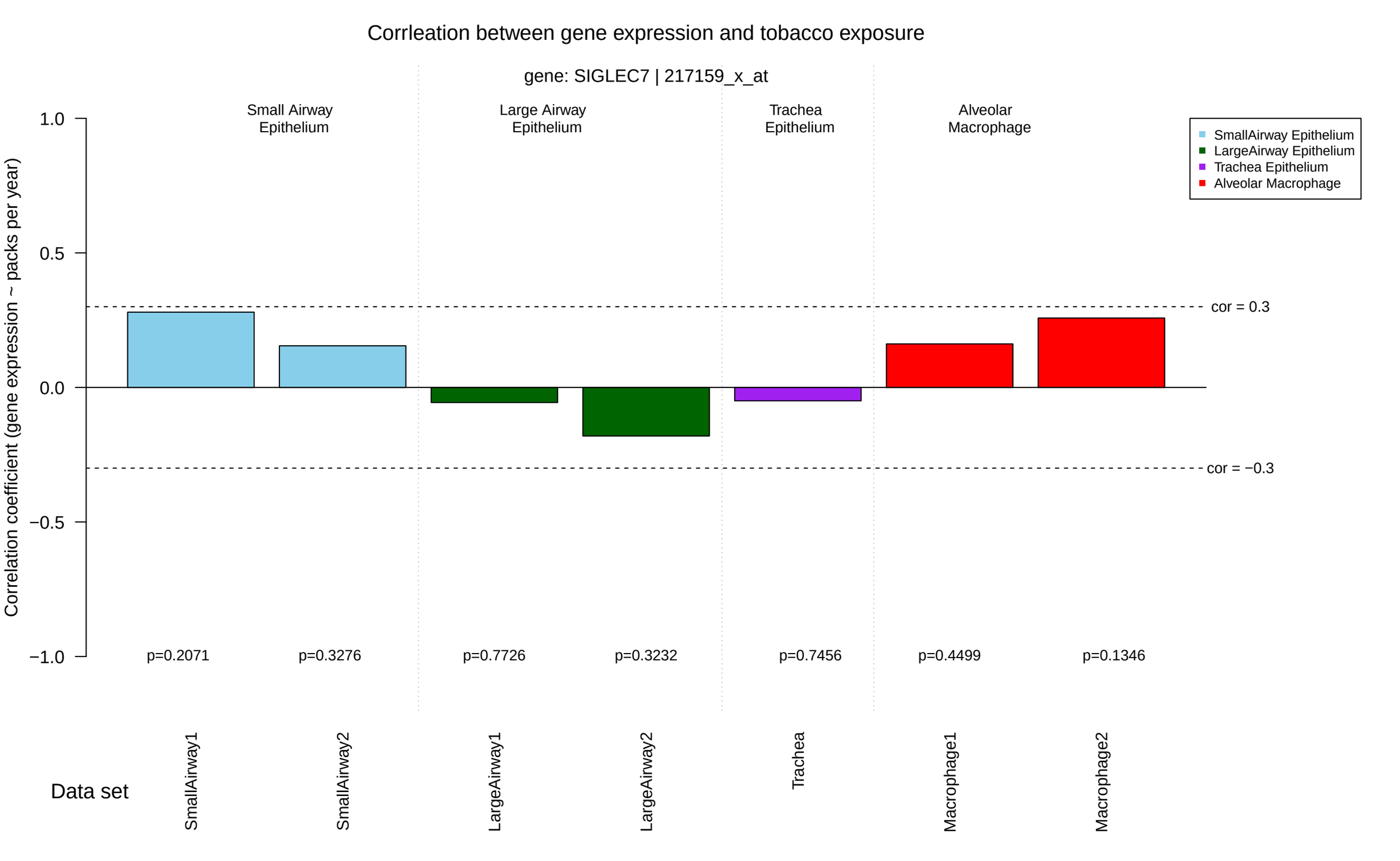

set: LargeAirway1 (p=0.0308)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9221)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5157)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7126)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0282)
SIGLEC7 | 217159_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC7 | 217159_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2071
p=0.3276
p=0.7726
p=0.3232
p=0.7456
p=0.4499
p=0.1346
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2